RT-DETR-Based Object Detection for Safety Equipment

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1. Domain/Dataset Choice, Sourcing & Labeling

Selected the Roboflow Universe *Construction Site Safety v2* dataset (CC BY 4.0), with 10,875 high-resolution construction scene images annotated across 12 safety equipment classes. Because the original dataset labeled equipment without annotating workers. Expanded the dataset to 13 classes by automatically labeling **Person** bounding boxes using a COCO-pretrained YOLO11 detector with a confidence threshold of ≥ 0.50. This yields 11,018 worker instances across 62.4% of images, enabling the model to reason over both equipment presence and worker context in a single unified detector.

2. Split Strategy & Justification

The dataset follows an **82.4% train (8,956 images) / 11.8% valid (1,279 images) / 5.8% test (640 images)** partition. Splits are strictly disjoint at the image level. Construction video frame sequences were kept contiguous within individual splits to prevent temporal leakage between training and evaluation. Stratification checks verified that rare classes (e.g. Foot protection, Respiratory protection) maintain consistent relative frequency across splits.

3. Quantitative Evaluation Metrics & Operational Meaning

Training stopped at **Epoch 71** (after 9h 40m on a Kaggle Tesla T4 GPU) as validation metrics showed clear saturation after peak fitness at **Epoch 65**. Final evaluation on validation and test splits:

Split / Metric	mAP@50	mAP@50-95	Precision	Recall	Key Class Drivers
Validation (1,279 imgs)	**79.90%**	**52.35%**	78.40%	79.49%	Peak fitness at Epoch 65; stopped at Epoch 71 (saturation)
Test Set (640 imgs)	**79.19%**	**50.99%**	78.05%	78.80%	Unseen test split generalization
• Head Protection (Hardhat)	80.27%	51.47%	86.19%	84.08%	Direct compliance driver; high precision
• No Head Protection (Bare)	82.82%	49.56%	81.21%	81.82%	Direct violation driver; catches 81.8% of unhelmeted heads
• Safety Vest / No Vest	74.92% / 63.01%	49.98% / 39.51%	76.67% / 77.99%	77.09% / 62.06%	High-contrast workwear causes vest confusion
• Foot Protection (Boots)	94.96%	74.13%	80.87%	93.88%	Strong ground-contrast features
• Person Class (Auto-labeled)	58.23%	48.62%	56.90%	61.03%	Occluded / truncated bodies limit person score

Operational Interpretation: Aggregate mAP@50 (79.2%) is heavily bolstered by distinct items like boots (95.0%) and eye protection (87.9%). However, for site safety auditability, the critical operational metric is the paired confusion between positive compliance and negative violation classes. Confusion matrix analysis reveals **41 instances of False Compliance on Head Protection** (bare head classified as hardhat) and **41 instances on Safety Vest**. In life-critical environments, False Compliance is dramatically more hazardous than False Alarm, directly establishing the requirement for Stage 3 confidence filtering.

4. Mid-Project Data Engineering Pivot

During baseline inspection, inspection detected that 4,990 annotation files (45.9% of the dataset) contained corrupted rows combining 5-parameter bounding box rows with raw polygon segmentation coordinates. Ultralytics silently dropped these images during training. Developing `fix_mixed_labels.py` cleansed corrupted lines, recovering 21,529 valid bounding boxes across all splits. Additionally, discovering that Roboflow lacked a Person class prompted developing an auto-labeling pipeline (`add_person_class.py`), preventing the need for a brittle two-stage cascaded detector at runtime.

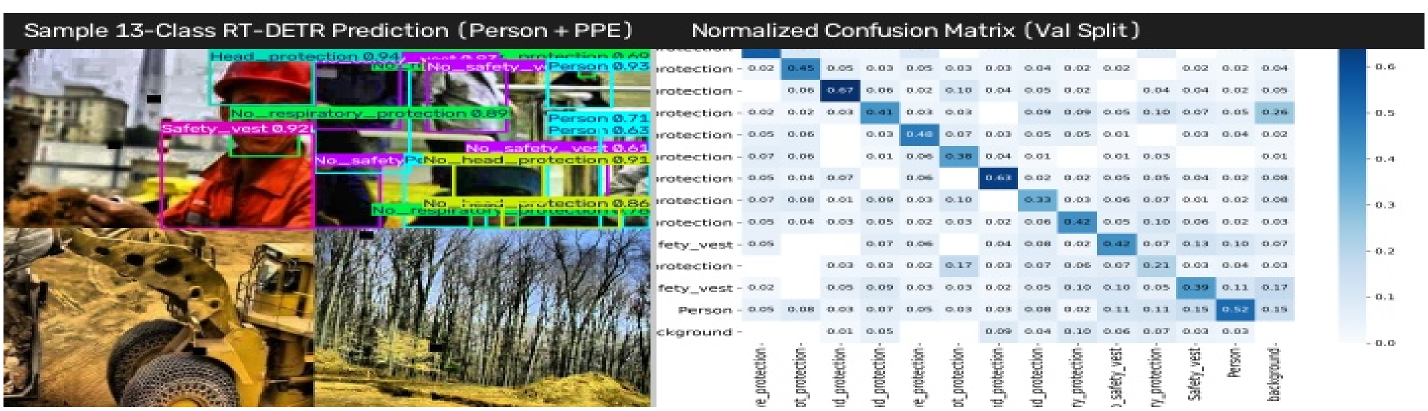
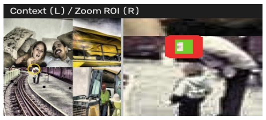
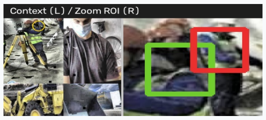
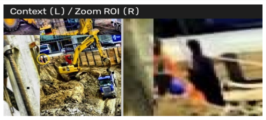
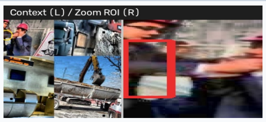


Figure 1: Left: Real-time RT-DETR-L detections demonstrating simultaneous Person anchor context and PPE bounding box predictions. Right: Normalized validation confusion matrix illustrating high on-diagonal accuracy and off-diagonal compliance confusions.

5. Systematic Failure Analysis (5 Real Test Cases)

Empirical inspection of the 640 held-out test predictions revealed five distinct structural failure modes:

Visual Evidence	Case / Test Image	Ground Truth vs Predicted	Root Cause & Safety Consequence
	**Case 1: Shadow False Compliance** `2008_008526...jpg`	GT: **No_head_protection** Pred: **Head_protection** (0.30)	Worker in shadow wearing dark beanie; brim curvature mimics hardhat dome. *Consequence: Dangerous false compliance clears unhelmeted worker.*
	**Case 2: High-Vis Workwear Confusion** `001425...7b40.jpg`	GT: **No_safety_vest** Pred: **Safety_vest** (0.86)	High-vis work shirt with tool straps mimics retroreflective vest tape. *Consequence: Safety vest violation missed by auditor.*
	**Case 3: Extreme Scale Disparity** `construction-3...148.jpg`	GT: **Hand_protection** & **Eye** Pred: *Zero detections* for PPE	Crane shot with workers < 80 px; gloves < 14 px collapse below stride-32 feature pyramid. *Consequence: Small PPE missed at distance.*
	**Case 4: Truncated Edge Silhouette** `-1680...73ce.jpg`	GT: **Person** + **Safety_vest** Pred: Vest (0.39), Person missed	Worker stepping into frame on left edge (65% cropped). Model misses worker silhouette and vest falls below 0.50 threshold.
	**Case 5: Equipment False Positive** `4c43875b...29c6.jpg`	GT: **Background (Mixer)** Pred: **Head_protection** (0.36)	Yellow convex hydraulic cap on mixer misclassified as helmet due to specular highlight and hemispherical geometry.

6. Hand-Written 3-Stage Decision Layer & Insufficient Information Case

To eliminate heavy framework overhead (no LangChain/CrewAI), `app/reasoning.py` implements an explicit 3-stage decision engine:

- **Deciding When to Call Detector vs. Not (Stage 1):** The Intent Router parses queries into `{needs_detection, query_type, target_class, negated}`. When queries ask about visual compliance or worker counts (e.g. *"How many workers have helmets?"*), `needs_detection=True`, triggering RT-DETR. When queries are off-topic or general knowledge (e.g. *"What is OSHA regulation 1926.100?"*), `needs_detection=False`, bypassing the vision detector entirely (`used_detection=False`) with zero latency.
- **Deterministic Reasoning (Stage 2):** Resolves counts, presence, and compliance pairings (`Head_protection` vs `No_head_protection`).
- **Confidence Guardrail (Stage 3):** Enforces strict safety thresholds to prevent ungrounded predictions.

Case Study: Real 'Insufficient Information' Production Execution

- **Test Image:** `-1680-_png_jpg.rf.73cee3e264b17ce5579750df4e4610f4.jpg` | **Query:** *"Is anyone wearing a safety vest?"*
- **Detector Output:** 1 detection → `Safety_vest` at `[0.0, 40.9, 22.8, 172.6]` with **conf = 0.3942** (border-cropped worker).
- **Decision Rule:** Stage 3 detects that vest confidence (0.394) is strictly below the safety threshold (0.50) and person context is unconfirmed.
- **Exact API Response Returned:** →

```json
{
  "answer": "I can't confidently answer this from
    the detections — confidence too low.",
  "confidence": "low",
  "used_detection": true
}
```

7. API Usage Instructions & Sample Payloads

Run Locally: `uv run uvicorn app.main:app --port 8000` | **Docker:** `docker run -d -p 8000:8000 rtdetr-ppe` | *Weights auto-download on first launch if missing.*

POST /detect (Object Detection & BBoxes)

Request:

```
curl -X POST "http://localhost:8000/detect?conf=0.25" \
  -F "file=@sample_site.jpg"
```

Response (200 OK):

```json
{
  "detections": [
    {"class": "Head_protection", "confidence": 0.9234,
     "box": [312.45, 84.12, 420.89, 195.67]},
    {"class": "No_safety_vest", "confidence": 0.8415,
     "box": [298.11, 190.54, 450.32, 480.21]}
  ],
  "image_width": 1280, "image_height": 720
}
```

POST /ask (Direct Grounded Safety Answer)

Request:

```
curl -X POST "http://localhost:8000/ask" \
  -F "file=@sample_site.jpg" \
  -F "question=How many workers are wearing hardhats?"
```

Response (200 OK):

```json
{
  "answer": "There are 2 workers wearing head protection.",
  "confidence": "high",
  "used_detection": true
}
```